Data-Driven Attitude Modeling and Control of Quadrotor UAVs using Koopman Operator

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***Abstract*—This paper proposes a data-driven approach to attitude modeling and control of the quadrotor UAV using Koopman operator theory. First, to address the limitations of traditional modeling methods, the Koopman operator is employed to learn the nonlinearity of the attitude dynamics from real data (i.e., angles, angular velocities and control torques). Then, the original nonlinear attitude system is reformulated in a high-dimensional linear attitude model. Second, a linear secondary regulator (LQR) control scheme is designed to ensures the stability of the attitude system. Finally, a physical platform of quadrotor attitude control is set up, experimental results validate the accurate prediction of the proposed modeling and control approach. A comparison with PID confirms the superiority of the proposed approach.**

***Index Terms*—data-driven, Koopman operator, attitude modeling, quadrotor, LQR control**

I. Introduction

In recent years, with the rapid development of electronic, communication, and energy technologies, quadrotor unmanned aerial vehicles (UAVs) have attracted widespread attention [1]–[3]. In the flight control of quadrotors, attitude control is a critical component in ensuring stable and reliable flight. However, quadrotors in practice are often affected by external environmental disturbances, model uncertainties, and measurement errors in internal parameters. These factors lead to unavoidable deviations between the theoretical model and actual models. Consequently, accurate attitude modeling and control is always an important issue in the field of control.

In the modeling of quadrotor UAVs, two main approaches are used, i.e., model-based methods and data-driven methods. The model-based methods relies on dynamics analysis. It uses physical laws to derive mathematical expressions of quadrotor dynamics. For example, the Newton Euler method builds a dynamic model by examining each force application point [4], [5]. The Lagrangian formulation uses generalized coordinates to represent positions and attitudes of all mass points, deriving overall dynamics from energy principles [6], [7]. However, model-based modeling is hard to describe complex dynamics due to the high coupling and nonlinearities. In addition, it also requires many simplifying assumptions (ignore friction fore, air resistance etc.), which reduce model accuracy.

The data-driven methods collects motion data from quadrotors, including system state and control input data, which offers clear benefits than model-based methods [8], [9]. For example, neural networks, machine learning, and statistical modeling methods establish a more accurate dynamic model [10]. This approach relies on observed data, showing strong generalization ability. It adapts to environmental changes and can predict system behavior accurately without an explicit mechanism. However, data-driven modeling has drawbacks, such as obtaining a suitable dataset for training proves difficult. The resulting models usually function as black boxes. How to ensure the systems' stability remains a challenge problem.

Koopman operator was first proposed by American mathematician Bernard O. Koopman in [11]. The main idea of Koopman operator is to map a nonlinear dynamical system into a high-dimensional or infinite-dimensional function space, enabling linear analysis of originally nonlinear evolution [12], [13]. This concept was later extended to broader dynamical system research and became a key tool in spectral analysis as well as data-driven methods. Extended Dynamic Mode

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Decomposition (EDMD) constructs a dictionary of basis functions to approximate the Koopman operator via least squares [15]. The EDMD output is then used to build a linear prediction model, after which standard control techniques design the system controller. This data-based linear approximation method ensures computational efficiency and improves control robustness. The Koopman operator provides a globally valid linear model of nonlinear dynamics, avoiding the local-only approximations and black-box nature of methods like neural networks. Furthermore, Koopman modes extracted via EDMD include both oscillation frequencies and decay rates [14].

For the attitude control problem of quadrotor UAVs. In [16]–[18], the stabilization and trajectory tracking of quadrotors is solved using linear schemes. These control methods are simple to implement. However, they neglect many factors, which limits control performance. In [19], [20], an MPC scheme is proposed, which requires online numerical optimization, resulting in high computational cost. Real-time performance is also constrained by hardware limits. It suits processes with strict constraints and relatively slow-changing conditions.

The main contributions of this paper are as follows:

1. Designing a data-driven attitude dynamic model for a quadrotor UAV based on Koopman operator theory.

2. Applying optimal control theory to achieve attitude tracking for the attitude dynamic model.

3. An experimental platform being built to validate the proposed method.

Next, Section II gives preliminaries knowledge. Section III proposes the modeling method using Koopman operator and the controller design. Section IV displays the experimental results. Section V concludes this paper.

II. Preliminaries

A. Traditional Attitude Model of Quadotor UAV

The quadrotor UAV is a homogeneous, symmetric rigid body subjected only to gravity and the lift forces generated by its propellers, its attitude dynamics equations can be derived from the Newton Euler equations [21] as

$$\begin{aligned} \dot{R} &= R\omega^{\times}, \\ J\dot{\omega} &= -\omega^{\times} J\omega + u + d(t), \end{aligned} \tag{1}$$

where $d(t) \in \mathbb{R}$ is the external disturbance, $R \in \mathbb{R}^{3\times3}$ is the rotation matrix from the body frame to the inertial frame, parameterized by roll ϕ, pitch θ, yaw ψ angles, $\omega = [\omega_1, \omega_2, \omega_3]^{\mathrm{T}} \in \mathbb{R}^3$ is the angular velocity expressed in the body frame, $\omega^{\times} \in \mathbb{R}^{3\times3}$ denotes the skew-symmetric matrix of the vector ω, and $J = \mathrm{diag}(J_{xx}, J_{yy}, J_{zz}) \in \mathbb{R}^{3\times3}$ denotes the moment of inertia matrix of the quadrotor UAV.

B. Koopman Operator Theory

The Koopman operator provides a way to represent nonlinear dynamical systems as linear operators acting on a space of observations [22]. Consider a discrete-time nonlinear system

$$x_{t+1} = f(x_t), \qquad x_t \in \mathcal{X} \subseteq \mathbb{R}^n. \tag{2}$$

where x_t is the state vector at t moment, $\mathcal{X} \subseteq \mathbb{R}^n$ is the state space. Rather than tracking the state x_t directly, we focus on scalar observations $\psi(\cdot) : \mathcal{X} \to \mathbb{C} \subseteq \mathbb{R}^m$, which assigns to each state x a complex number $\psi(x)$. The Koopman operator $\mathcal{K} : \mathcal{F} \to \mathcal{F}$ is defined by

$$\psi(x_{t+1}) = \psi\big(f(x_t)\big) = \mathcal{K}\,\psi(x_t)\,, \quad \psi \in \mathcal{F}. \tag{3}$$

Thus, $\mathcal{K}$ advances each observable $\psi(x)$ along the flow generated by $f(\cdot)$.

In practice, using data-driven methods such as Extended Dynamic Mode Decomposition (EDMD) can yield a tractable, finite-dimensional linear model that captures the essential behavior of the original nonlinear system.

III. Linear Predictors for Quadrotor

In this section, a data-driven attitude modeling approach of quadrotor UAVs using Koopman operator is proposed and a LQR control scheme is designed to ensure the stability of the attitude system.

A. Data-Driven Attitude Model of Quadotor UAV

Theorem 1: For attitude system (1), if angles ϕ, θ, ψ, angular velocities $\omega \in \mathbb{R}^3$ and control inputs $u \in \mathbb{R}^3$ are measured, then there exists state $x = [R, \omega^{\times}] \in \mathbb{R}^{3\times6}$, and a lift map $\Phi(\cdot)$ is selected as, for $N \in \mathbb{N}$ and $N \geq 2$,

$$\begin{aligned} \Phi(x) &= [\xi_1(x), \xi_2(x), \xi_3(x), \xi_4(x), ..., \xi_N(x)]^{\mathrm{T}}, \\ &= [\underline{R}^{\mathrm{T}}, \underline{R\omega^{\times 1}}^{\mathrm{T}}, \underline{R\omega^{\times 2}}^{\mathrm{T}}, \cdots, \underline{R\omega^{\times (N-1)}}^{\mathrm{T}}], \end{aligned} \tag{4}$$

where $(\cdot)^m (m = 1, 2, \cdots, N-1)$ represents the m power of variable, $\underline{(\cdot)} = vec(\cdot)$ represents the vectorization of the matrix. For a matrix $A = [a_{ij}] \in \mathbb{R}^{3\times3}$, there exists

$$\underline{A} = [a_{11}, a_{12}, a_{13}, a_{21}, a_{22}, a_{23}, a_{31}, a_{32}, a_{33}]^{\mathrm{T}}. \tag{5}$$

Based on (4), an approximate linearization model of system (1) can be obtained as

$$\begin{aligned} \Phi(x_{k+1}) &= A\,\Phi(x_k) + B\,u_k \\ &:= K \begin{bmatrix} \Phi(x_k) \\ u_k \end{bmatrix}, \end{aligned} \tag{6}$$

where $A, B, K = [A, B]^{\mathrm{T}}$ are proper constant matrices, and x_k, u_k are the system state and control input at time kT_s, T_s is the sampling period, $k = 0, 1, 2, 3, \cdots$.

Proof 1: According to [23], the lift state for the attitude system (1) is chosen as

$$\Gamma = [\underline{R}^{\mathrm{T}}, \underline{z_1}^{\mathrm{T}}, \underline{z_2}^{\mathrm{T}}, \underline{z_3}^{\mathrm{T}}, ...]^{\mathrm{T}}, \tag{7}$$

where $z_i = R\omega^{\times i}$ $(i = 1, 2, 3, \cdots)$, then it can be written as

$$\dot{\Gamma} = C\Gamma + D\bar{u}. \tag{8}$$

where $\bar{u} = -\omega^{\times} J\omega + u + d(t)$ and C, D is the coefficient matrix.

System states and control inputs are measured and captured in the $M+1$ moments, which establishes a data set

$$Data = \{(x_k, u_k) | k = 1, 2, ..., M+1\}, M \in \mathbb{R}, \tag{9}$$

where k denotes the time kT_s and T_s is the sampling period. Then, $P \in \mathbb{R}^{(9N+3)\times M}$ and $Q \in \mathbb{R}^{9N\times M}$ is given as

$$P = \begin{bmatrix} \Phi(x_1) & \Phi(x_2) & \dots & \Phi(x_M) \\ u_1 & u_2 & \dots & u_M \end{bmatrix}$$
$$Q = [\Phi(x_2), \Phi(x_3), ..., \Phi(x_{M+1})], \tag{10}$$

Thus, K is the solution of the problem as follows

$$\begin{aligned} K &= \arg\min_K \left\|Q - KP\right\|_2^2, \\ &= \arg\min_K \sum_{i=1}^{M} \left\|\Phi(x_{i+1}) - K[\Phi(x_i), u_i]^{\mathrm{T}}\right\|_2^2. \end{aligned} \tag{11}$$

Using the data matrices, it can be obtained that

$$\begin{aligned} G_1 &= \frac{1}{M}\sum_{i=1}^{M} \Phi(x_{i+1})[\Phi(x_i), u_i]^{\mathrm{T}}, \\ G_2 &= \frac{1}{M}\sum_{i=1}^{M} [\Phi(x_i), u_i][\Phi(x_i), u_i]^{\mathrm{T}}, \end{aligned} \tag{12}$$

so that the solution of the problem is

$$K = G_1 {G_2}^{\dagger}, \tag{13}$$

where ${G_2}^{\dagger}$ is the Moore Penrose pseudoinverse. K may then be partitioned into two blocks, which is

$$K = [A \quad B], \tag{14}$$

B. Design of LQR controller for Data-Driven Attitude Model

For Koopman operator linear system (6), if the feedback control input is chosen as

$$u_k = -\Omega_{LQR}\Phi(x_k), \tag{15}$$

and the quadratic cost function is designed to obtain the minimum value, given by

$$J \;=\; \sum_{k=0}^{\infty}\big(\Phi(x_k)^{\mathrm{T}}\, Q\, \Phi(x_k) \;+\; u_k^{\mathrm{T}}\, R\, u_k\big). \tag{16}$$

where

$$\begin{aligned} \bar{Q} &= \begin{bmatrix} Q & 0 \\ 0 & 0 \end{bmatrix} \in \mathbb{R}^{9N\times 9N}, \\ Q = Q^{\mathrm{T}} &\geq 0, \qquad R = R^{\mathrm{T}} > 0. \end{aligned} \tag{17}$$

then the optimal control of system system (6) is achieved.

If pair (A, B) is controllable, then defined the discrete algebraic Riccati equation as follows

$$P \;=\; A^{\mathrm{T}} P\, A - A^{\mathrm{T}} P\, B\,(R{+}B^{\mathrm{T}} P\, B)^{-1} B^{\mathrm{T}} P\, A + \bar{Q}. \tag{18}$$

This equation has a unique symmetric positive semi-definite solution which can be defined as the optimal gain

$$\Omega_{LQR} = (R + B^{\mathrm{T}}PB)^{-1}B^{\mathrm{T}}PA. \tag{19}$$

IV. Experiment Result

A. Experimental Platform

The attitude platform of quadrotor UAV is an embedded real-time physical simulation platform, primarily consisting of two parts: the attitude platform and the ground station. The attitude platform includes components such as the support chassis, slip rings, and the quadrotor UAV body, while the UAV body is composed of the frame, controller, and power unit. The flight controller uses the STM32F429 chip as the main processor, with a 180 MHz clock speed providing efficient computing power and quick response. The motors are SUNNY SKY X2212 brush less motors with a KV value of 1400, paired with 9-inch APC propellers as the power setup.

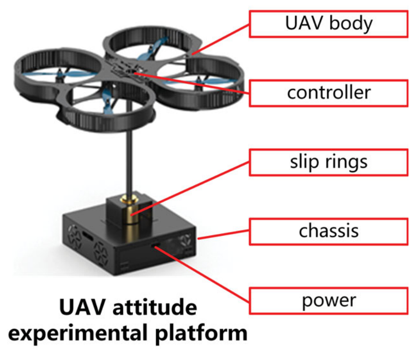

Fig. 1. UAV attitude experimental platform components

The ground station has the functions of real-time interaction and loading control algorithms. It communicates in real-time with the quadrotor UAV, sending commands to set the desired attitude, receiving current attitude information, and displaying data and waveforms in real-time on the interface. The real-time attitude is also visualized through a 3D model.

B. Experiment Results of Attitude Modeling

As for selecting the lifting order N, we found that simply increasing N (and hence the model dimension) does not guarantee better accuracy. When N is too large, the proliferation of nonlinear observations leads to substantial regression error; when N is too small, there are too few observations to capture the true dynamics. To determine the optimal N, we generated attitude trajectories from the governing dynamics, fed them into the lifted model described above, and compared the true versus predicted values for various N. From Table I, it is observed that $N = 5$ yields the smallest prediction error. With $N = 5$ the lifted system has dimension 45. In order to recover the original state from the 45-dimensional lifted state for simulation and comparison, we define $H = [I_{18\times18}, 0_{18\times27}]$. It then follows that:

$$x_k = Hz_k \tag{20}$$

Using the error at $N = 8$ as a reference value of 1, the relative prediction errors for different values of N are summarized in Table I.

To evaluate the practical performance of the Koopman operator, we carried out experiments on a custom quadrotor

TATBLE I. RELATIVE PREDICTION ERRORS FOR DIFFERENT N

N	3	4	5	6	7	8(base)
Relative error(%)	0.67	0.64	0.58	0.61	0.72	1.00

attitude platform. We generate motion trajectories by commanding time-varying reference attitudes. To ensure sufficient excitation in the measured angular velocities, the reference attitude follows random sinusoidal signals with amplitudes up to $\pm 10^{\circ}$ and frequencies below 0.6Hz. Forty trajectories were recorded, each $15s$ long and sampled every $0.05s$. At each sample we logged the quadrotors attitude angles, angular velocities and the implicit control torques. In the Forty random attitude trajectories, thirty were used to train the Koopman linear model and ten were held out for validation. For one representative validation trajectory, Fig. 2 and Fig. 3 plot the measured versus predicted body-fixed attitude angles and angular velocities, demonstrating the models ability to capture the quadrotors nonlinear dynamics.

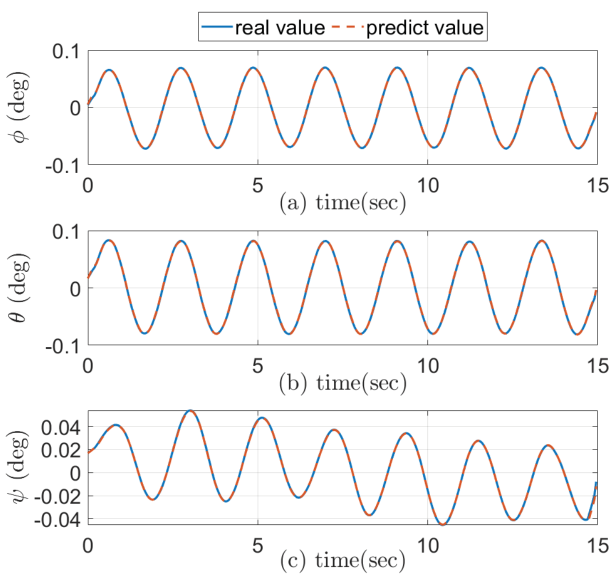

Fig. 2. Attitude prediction under the Koopman model

C. Experiment Results of Koopman-LQR Control

Using LQR control theory to compute the required control input u_k. The desired attitude angles are set to zero. Fig. 4 demonstrates accurate tracking of the reference curve. In the ϕ and θ channels, the algorithm proposed in this paper achieves both shorter adjust time and smaller overshoot compared to the PID algorithm. In the ψ channel, it exhibits a faster convergence speed during the initial stage, the detailed data analysis is shown in Table II. The proposed LQR algorithm outperforms the PID controller in settling time as well as steady-state error. Fig. 5 depicts the systems attitude angular velocity and Fig. 6 depicts the control torque.

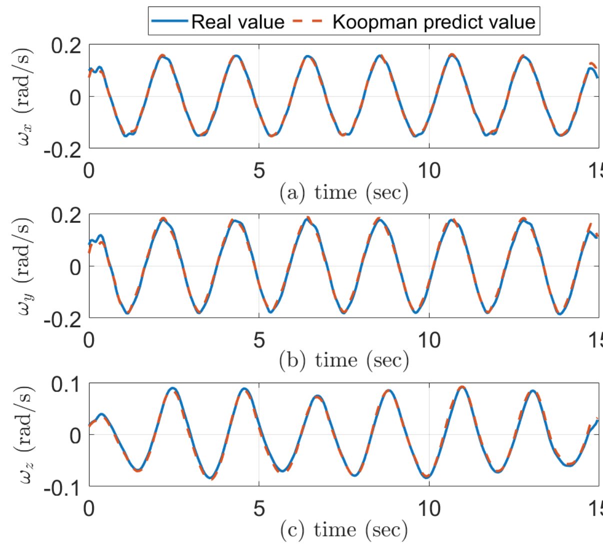

Fig. 3. Attitude velocity prediction under the Koopman model

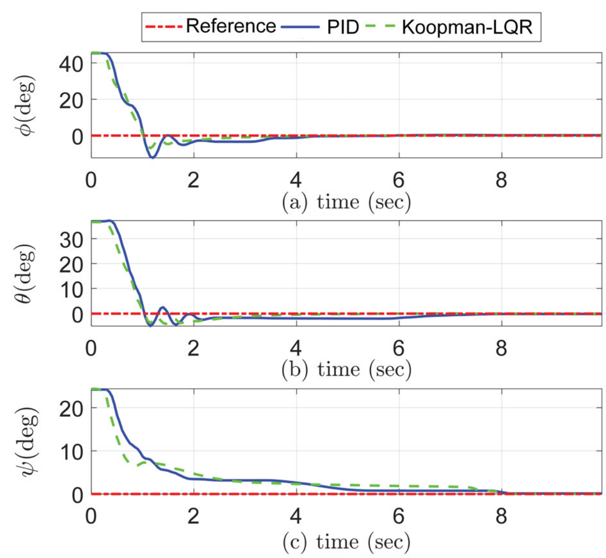

Fig. 4. Attitude response under PID controller and Koopman-LQR controller

TATLBE II. COMPARISON OF DYNAMIC PERFORMANCE UNDER PID CONTROLLER AND KOOPMAN-LQR CONTROLLER CONTROLLER

Channel	Algorithm	Settling time	Overshoot
ϕ	PID	3.40s	27.00%
	Koopman-LQR	2.05s	15.38%
θ	PID	5.95s	13.55%
	Koopman-LQR	2.60s	10.28%
ψ	PID	7.50s	-
	Koopman-LQR	4.85s	-

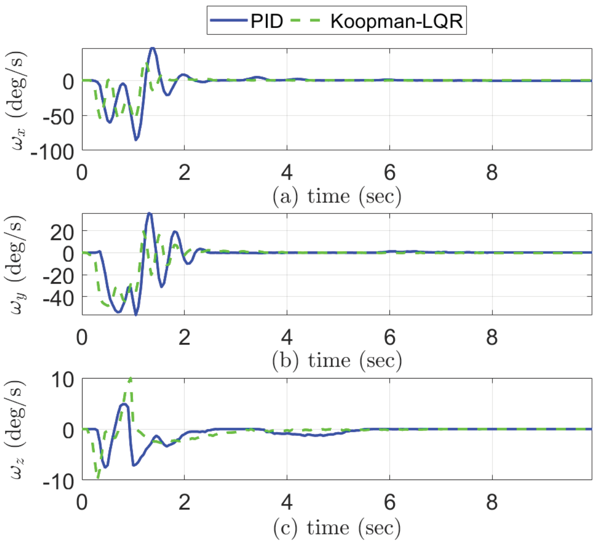

Fig. 5. Attitude velocity response under PID controller and Koopman-LQR controller

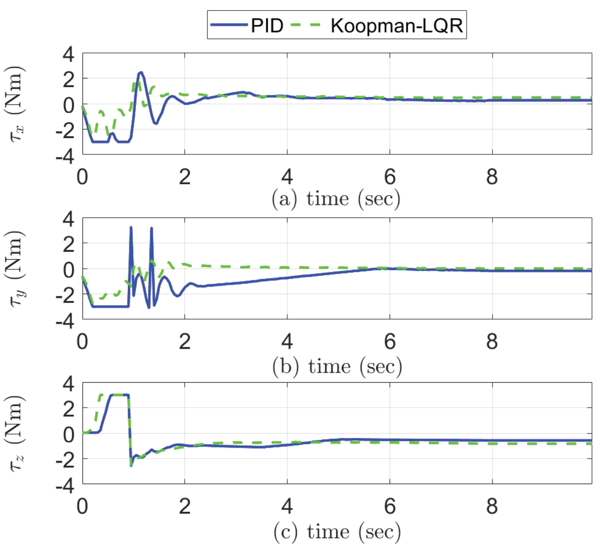

Fig. 6. Attitude control torque under PID controller and Koopman-LQR controller

V. Conclusion

This paper proposes a modeling method and LQR control approach for the attitude control of quadrotor UAVs based on the data-driven approach. Using Koopman operator, an attitude model for the quadrotor UAV is established, revealing the relationship between its inputs and outputs. Subsequently, an LQR controller is designed based on this model, and it is proven that the system is asymptotically stable. Finally, the algorithm is validated and compared with other performance methods on the physical quadrotor UAV attitude control platform. Experimental results show that the proposed method has significant advantages. In the future, the method proposed will be extended to the position model of quadrotor UAV and other types of robots.

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